Cyclistic Bike-Share: Data-Driven Conversion Strategy Understanding User Behavior to Drive High-Margin Membership Growth

Presented by: Pritesh Das (Junior Data Analyst, Marketing Analytics Team, Cyclistic)

The Mission & Business Task

<u>Shifting Focus from Broad Awareness to High-Margin Conversion:</u>

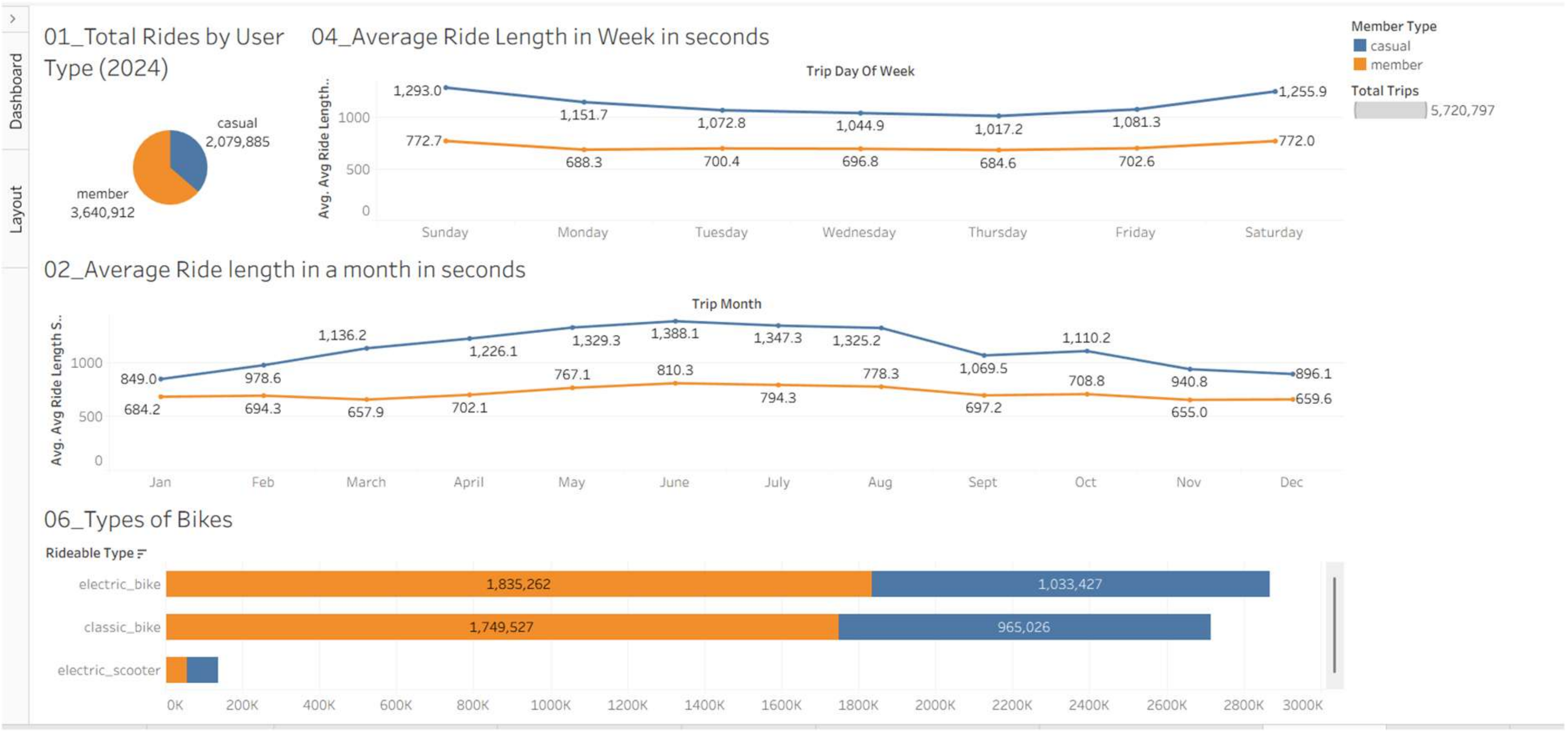
- The Executive Mandate: The Director of Marketing, Lily Moreno, has identified that Cyclistic's future success depends entirely on maximizing the number of annual memberships.
- **The Strategic Shift:** Moving away from broad brand awareness campaigns to target **casual riders** (single/full-day pass holders) who are already familiar with our service and represent a high-potential segment for growth.
- **The Core Analytical Question:** *"How do annual members and casual riders use Cyclistic bikes differently?“*
- **The Goal:** Deliver compelling, data-backed insights and visualizations to design a marketing playbook approved by the executive team.

⚙️ Data Pipeline & Technical Rigor

Building an Airtight Foundation for Strategic Decisions:

- **Data Scale:** Analyzed a massive public dataset encompassing a full calendar year of historical trip logs (January 2024 to December 2024).
- **Advanced Processing:** Utilized **SQL Server Management Studio (SSMS 22)** to check schema consistency and execute UNION ALL consolidation across 12 monthly transaction files.
- **Data Cleansing & Integrity:** Systematically isolated and removed **1,496,735 rows** containing null or missing values and Engineered new time-delta features: avg_ride_length_in_seconds, day_of_week, and month.
- **Anomalous Outlier Filtration:** Excluded trips under 1 minute (false starts/docking adjustments) and over 24 hours (lost/unreturned equipment) to eliminate metric skew.
- **Final Verified Audited Dataset: 4,224,062 clean rows** optimized for analytics.

1st Dashboard



Key Insights — Volume, Duration, and Fleet Preferences (1st Dashboard)

Total Rides and Market Share

- Annual members drive the core volume, **completing 3,640,912 trips, which accounts for 63.64% of total ridership.**
- Casual riders account for a strong secondary market share of **36.36%, executing 2,079,885 trips.**

Weekly Ride Duration Trends

- Casual riders consistently cycle for longer durations than members every single day of the week.
- Casual trip duration peaks sharply on **Sundays at 1,293 seconds (21.55 minutes) and Saturdays at 1,255.9 seconds (20.93 minutes).**
- Member trip duration remains incredibly flat and steady, peaking slightly on **Saturdays at 772 seconds (12.86 minutes) and Sundays at 772.7 seconds (12.88 minutes).**

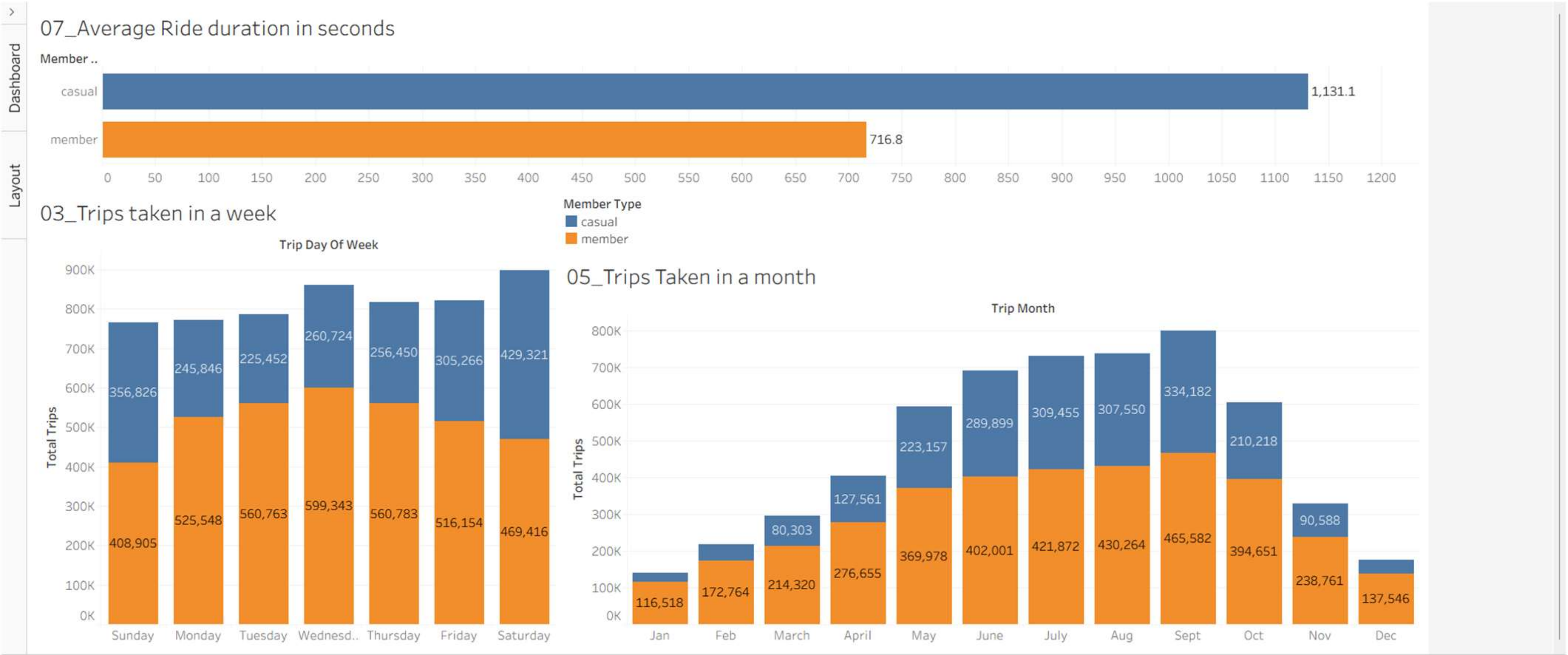
Monthly Duration and Seasonality

- Both user segments experience their longest individual trip lengths in June, **with casual riders reaching an average maximum of 1,388.1 seconds (23.13 minutes) and members peaking at 810.3 seconds (13.50 minutes).**
- Trip lengths for both groups decline significantly during winter, bottoming out around November and December.

Fleet and Asset Preferences

- Both customer cohorts show a clear preference for electric bicycles over classic, traditional models.
- For electric bikes, **annual members logged 1,835,262 trips while casual riders completed 1,033,427 trips.**
- For classic bikes, **annual members logged 1,749,527 trips compared to 965,026 trips by casual riders.**
- Electric scooters represent a notable micromobility niche, where casual riders out-utilize members.

2nd Dashboard



Key Insights — Ride Duration, Weekly Volume, and Monthly Trends (2nd Dashboard)

Overall Average Ride Duration

- Casual riders display significantly higher individual asset engagement, **averaging 1,131.1 seconds (~18.85 mins) per trip.**
- Annual member rides **average 716.8 seconds (~11.94 mins)**, indicating a point-to-point utility commute pattern.

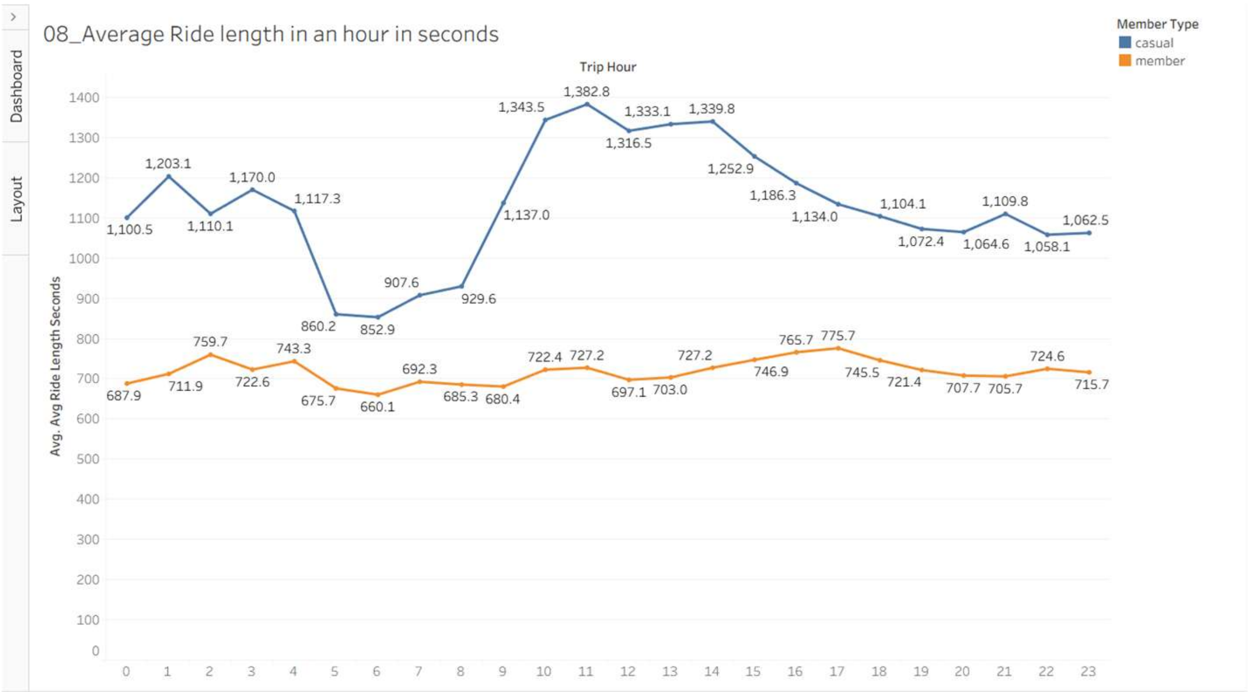
Weekly Ridership Volume

- Annual members provide consistent, high-volume fleet activity throughout the standard workweek, **peaking sharply on Wednesdays with 599,343 rides.**
- Member activity drops to its **lowest weekly level on Sundays with 408,905 rides**.
- Casual riders show a distinct preference for weekend usage, surging to their **highest weekly volume on Saturdays with 429,321 rides.**
- Casual activity drops to its **lowest weekly point on Tuesdays with 225,452 rides.**

Monthly Volumetric Trends

- Both customer segments experience their lowest service utilization in January, **with annual members at 116,518 rides and casual riders at 23,642 rides.**
- Ridership increases steadily during the warmer spring and summer months for both groups.
- Casual riders reach their **absolute highest peak of the year in September with 334,182 rides.**
- Annual members **maximize their total monthly volume in July with 465,582 rides.**

3rd Dashboard



Key Insights — Hourly Ride Length Dynamics

Casual User Hourly Behavior

- Casual riders maintain a significantly higher average trip duration than members across every single hour of the day.
- Casual riders record their absolute longest average rides during the midday window, peaking at **11:00 AM with an average of 1,382.8 seconds (~23.05 minutes).**
- A secondary high-duration plateau is sustained for casual riders throughout the afternoon, **remaining above 1,300 seconds from 10:00 AM through 2:00 PM (14:00).**
- The shortest average trips for casual riders occur in the early morning, **hitting a low of 852.9 seconds (~14.22 minutes) at 6:00 AM.**

Annual Member Hourly Behavior

- Annual members exhibit a highly stable and flat trip duration baseline, **remaining between 660 and 780 seconds throughout the entire 24-hour cycle.**
- Members reach their maximum average trip duration during the evening commute window, **peaking at 5:00 PM (17:00) with 775.7 seconds (~12.93 minutes).**
- Similar to casual riders, members record their shortest average trip durations in the early morning, hitting a floor of 660.1 seconds (11.00 minutes) at 6:00 AM.

Comparative Takeaways

- **At 11:00 AM**, the casual ride duration gap reaches its widest margin, where casual trips are **nearly double the length of member trips (1,382.8 seconds vs. 727.2 seconds).**
- **The synchronized 6:00 AM drop for both segments** suggests that early morning riders—regardless of membership status—primarily prioritize fast, direct, and efficient point-to-point travel.

🚀 Strategic Playbook Part 1 — Product & Packaging

Meeting Casual Riders Where Their Habits Are:

1. The "Summer Explorer" Tier (Membership Personalization)

- ***The Insight:*** Casual use spikes dramatically in summer for long, continuous durations.
- ***The Strategy:*** Introduce custom, short-term programmatic subscriptions (monthly passes or seasonal packages) to bridge the financial gap for riders who aren't ready for a 12-month commitment but demand summer volume.

2. Electric Fleet Incentives

- ***The Insight:*** Casual users show an outsized preference for electric bikes and scooters.
- ***The Strategy:*** Introduce member-exclusive benefits for electric fleet options, such as priority app unlocking, free advanced reservations, or lowered per-minute rates, converting asset affinity into membership sign-ups.

🚀 Strategic Playbook Part 2 — Engagement & Marketing

Data-Timed Digital Outreach & Social Value

3. The "Weekend Warrior" Loyalty Infrastructure

- ***The Insight:*** Casual riders hit maximum volume on Saturdays and Sundays.
- ***The Strategy:*** Launch a points-based loyalty program where frequent weekend single-pass and full-day riders automatically accumulate points that can be converted directly into a currency discount toward purchasing a standard annual subscription.

4. Hyper-Targeted Digital Campaigns

- ***The Insight:*** Casual users peak heavily on summer weekends around 11:00 AM.
- ***The Strategy:*** Trigger geotargeted digital ads on social media platforms precisely during peak casual windows (Weekends at 11:00 AM, May–September), utilizing user-generated content emphasizing premium lifestyle and recreation.

5. Member-Exclusive Experiential Events

- ***The Strategy:*** Organize exclusive summer community group rides, neighborhood navigation challenges, and urban exploration events to build a strong community aspect around owning a Cyclistic membership.

Conclusion: The Bottom Line

Transforming Leisure into High-Margin Retained Revenue

1. **The Core Contrast Uncovered:**

- **Annual Members:** Utilize Cyclistic as an essential point-to-point **utility** for occupational commuting.
- **Casual Riders:** Leverage the system as a premium **leisure and experiential** asset.

2. **Strategic ROI:** Transitioning from broad, generic brand awareness to precise behavioral conversion unlocks a predictable, compounding revenue stream.

3. **The Next Step:** Implementing these tailored digital strategies, loyalty hooks, and customized packages will successfully convert casual riders into high-margin annual members, ensuring the long-term profitability and sustainable growth of Cyclistic.

“Thank You”